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ORIGINAL RESEARCH ARTICLE

Acute Effects of *Ad Libitum* Use of Commercially Available Cannabis Products on the Subjective Experience of Aerobic Exercise: A Crossover Study

Laurel P. Gibson[1] · Gregory R. Giordano[1] · L. Cinnamon Bidwell[1] · Kent E. Hutchison[2] · Angela D. Bryan[1]

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Abstract

Objective The present study aimed to examine the acute effects of legal-market cannabis on regular cannabis users' subjective responses to exercise in a controlled laboratory environment.

Background Given the stereotype that cannabis is associated with extreme sedentary behavior, there are concerns that cannabis legalization may exacerbate the US physical inactivity epidemic. However, despite these concerns, recent years have seen considerable public interest in the use of cannabis *concurrently* with exercise (e.g., running).

Methods The present study compared participants' experiences of exercise without cannabis to their experiences of exercise after acute ad libitum use of one of two commercially available cannabis flower products: a Δ9-tetrahydrocannabinol-dominant or a cannabidiol-dominant product. Participants ($N=42$) were regular cannabis users between the ages of 21 and 39 years (mean = 30.81 years, standard deviation = 4.72 years).

Results Although participants reported a more positive affect ($p<0.001$), enjoyment ($p<0.001$), and runner's high symptoms ($p<0.001$) during their cannabis (vs non-cannabis) exercise appointment, they also reported more exertion ($p=0.04$). Pain levels were very low and did not differ between appointments ($p=0.45$). Effects appeared to depend, in part, on cannabinoid content; there was a larger difference in enjoyment ($p=0.02$), and a smaller difference in exertion ($p=0.02$), between the cannabis and non-cannabis exercise appointments among participants in the cannabidiol (vs Δ9-tetrahydrocannabinol) condition.

Conclusions To our knowledge, this is the first study to investigate the acute effects of commercially available cannabis on subjective responses to exercise in a laboratory environment. Our findings suggest that, among regular cannabis users who use cannabis in combination with exercise, cannabis use prior to exercise may lead to increases in both positive and negative aspects of the subjective exercise experience. Research using diverse samples, exercise modalities, and methodologies (e.g., placebo-controlled trials) is needed to establish the generalizability of these findings.

1 Introduction

Legislation surrounding the medical and non-medical use of cannabis has undergone significant changes in the USA in recent years. The majority of US adults now have access to medical cannabis in some capacity, and a growing percentage of US adults, almost 50%, now live in adult-use cannabis states. Coinciding with this rapidly evolving legal landscape, the USA has seen an increase in the use [1, 2] and

Key Points

In a sample of individuals who use cannabis with exercise, participants reported higher levels of positive mood states (e.g., enjoyment) when exercising under the influence of cannabis, regardless of whether they used a Δ9-tetrahydrocannabinol-dominant or cannabidiol-dominant product.

At the same time, participants who used a Δ9-tetrahydrocannabinol-dominant product indicated that exercise felt more effortful after cannabis use.

These findings suggest that, among individuals who use cannabis with exercise, acute cannabis use may have both positive and negative impacts on the exercise experience.

✉ Laurel P. Gibson
laurel.gibson@colorado.edu

1 Department of Psychology and Neuroscience, University of Colorado Boulder, Boulder, CO, USA

2 Department of Psychiatry, University of Colorado Anschutz Medical Campus, Aurora, CO, USA

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acceptance of cannabis [3, 4]. As such, it is imperative that research investigates the public health implications of cannabis legalization, including its impact on physical inactivity, a major risk factor for obesity and noncommunicable diseases.

Given the popular stereotype that cannabis is associated with extreme physical inactivity, or "couch lock," there are concerns that cannabis legalization may exacerbate the US physical inactivity epidemic [5]. Yet despite these concerns, several cross-sectional studies have found that cannabis users engage in more exercise than non-users [6–11]. Moreover, in recent years, there has been considerable public interest in the use of cannabis *concurrently* with exercise [12–14], despite limited empirical evidence to support this behavior. Indeed, the use of cannabis with various forms of exercise, most notably running, has been advocated for by both recreational and professional athletes [15–19]. These anecdotal reports of cannabis use with exercise, in tandem with cross-sectional associations between cannabis use and physical activity, underscore the need for more research aimed at elucidating the *acute* effects of cannabis on exercise.

To date, there is a dearth of research examining the acute effects of cannabis on exercise. As previously reviewed [21, 22], the published literature is limited to a small number of studies that focus exclusively on exercise physiology or exercise and sports *performance*, with some studies documenting an ergolytic effect of acute cannabis use on performance [23–26], and others documenting no effect [27, 28]. It is important to note, however, that with one exception [28], these studies were published over 30 years ago and utilized cannabis forms and potencies that are not commonly available or used in state-regulated markets today (e.g., low Δ9-tetrahydrocannabinol [THC] concentrations), underscoring the need for ecologically valid research designs utilizing commercially available cannabis products that reflect today's legal market. Moreover, most individuals using cannabis in combination with exercise report doing so not for performance-related reasons, but rather to facilitate a more positive response to exercise [22]. For instance, two recent survey studies of adults who use cannabis in combination with exercise found that although most participants agreed that cannabis use did *not* improve their exercise performance, most agreed that cannabis increased their enjoyment of exercise [10, 29]. This underscores the need for research aimed at understanding the acute effects of cannabis on the *subjective* experience of exercise, rather than performance alone.

Importantly, only one study, to our knowledge, has moved beyond survey methodology to examine the acute effects of cannabis on the subjective experience of exercise [30]. In this observational study, participants were instructed to complete two separate runs, one run without cannabis (non-cannabis run) and one run after ad libitum use of a legal-market cannabis product of their choice (cannabis run), at a time and outside location convenient to them. Participants reported experiencing similar levels of perceived exertion, less pain, and more positive affect, enjoyment, tranquility, and runner's high symptoms during their cannabis run. Of note, although participants ran the same route for their cannabis and non-cannabis runs, their running pace was free to vary, with participants running an average of 19 s/km slower during their cannabis run. As such, it is unclear whether these findings hold in a more controlled environment.

Given reported effects of cannabis on the positive subjective experience of exercise, it is interesting to note that acute exercise has been shown to activate the endocannabinoid system by increasing the release and circulation of endocannabinoids (eCBs) [31–34]. In human studies, these exercise-induced eCB increases have been associated with increases in positive mood outcomes [33, 34], as well as reductions in depression, tension, and pain immediately following exercise [32, 34]. Moreover, emerging evidence suggests that circulating eCBs, rather than endorphins, may be responsible for the feelings of euphoria, relaxation, and effortlessness occasionally experienced during intense or lengthy bouts of aerobic exercise (c.f., "runner's high") [35, 36]. As exogenous cannabinoids (e.g., THC, cannabidiol [CBD]) interact with the eCB signaling system [37], it is possible that exogenous cannabinoids may mirror the effects of eCBs in the context of exercise. Indeed, at certain doses, acute cannabis use, particularly of chemovars high in THC, has been associated with increases in euphoria [38] and positive affect [39], raising the possibility that acute cannabis use may improve an individual's affective response to exercise. To the extent that acute cannabis use facilitates a more positive affective response to exercise, it may also lead to increases in exercise enjoyment and reductions in perceived exertion, as exercise-induced positive affect has been linked with both higher levels of enjoyment [40] and lower levels of exertion [41] during exercise. Further, as emerging evidence suggests that THC, CBD, and their combined use may have both analgesic and anti-inflammatory properties [42], acute cannabis use may mitigate exercise-induced pain. However, it is important to acknowledge that the subjective effects of cannabis are by no means universally positive; acute cannabis use has been associated with tiredness [43], paranoia [44], and anxiety [45], and as such, cannabis use could also negatively impact the subjective experience of exercise.

1.1 The Present Study

The present study investigated the acute effects of legal-market cannabis products on regular cannabis users' subjective responses to exercise in a controlled laboratory environment.

Using a 2 (within: exercise appointment) × 2 (between: cannabis condition) mixed design, we compared exercise without cannabis to exercise after ad libitum use of one of two commercially available cannabis products: a THC-dominant flower product or a CBD-dominant flower product. The first aim of the study was to examine the acute effects of using versus not using cannabis on ratings of perceived exertion (RPE), pain, affective valence, affective arousal, enjoyment, and runner's high symptoms during a 30-min exercise bout (within-subjects). Specifically, we hypothesized that participants would report (1) lower levels of RPE and pain and (2) higher levels of positive affective valence, affective arousal, enjoyment, and runner's high symptoms during the exercise bout where they used cannabis compared with the exercise bout where they did not use cannabis. The second aim was to examine the acute effects of using a THC-dominant versus CBD-dominant flower product on the same outcomes (between-subjects). Given limited research on the effects of cannabinoid content on responses to exercise, we made no hypotheses regarding the direction of these effects. Additionally, exploratory analyses examined the acute effects of cannabis use on resting heart rate and percentage of maximal heart rate during the exercise bout.

2 Methods

2.1 Participants

Participants were recruited from the greater Boulder area using advertisements posted online and in community public spaces. Interested participants were screened via telephone by trained research staff, and were eligible to participate if they (1) were aged 21–40 (male) years or 21–50 (female) years (i.e., 5 years below the age at which an individual's risk of heart attack begins to increase [46]), (2) had used cannabis while running or jogging at least three times in the past 12 months or at least six times in their lifetime with no negative effects, (3) were physically capable of safely engaging in moderate-to-vigorous intensity exercise for 30 or more minutes, and (4) lived within a 20-min drive from the exercise facility. Participants were screened to exclude those with cardiovascular or respiratory illnesses; a close family history of an adverse cardiac event before the age of 50 years; high blood pressure; metabolic disorders; a muscle, joint, or bone problem or an injury that limits movement, makes exercise painful, or could be made worse by exercising; psychotic symptoms or bipolar disorder; and those reporting a serious medical illness or injury in the past 6 months. Additionally, if female, participants could not be pregnant or nursing.

Power was estimated using G*Power [47] and was based on the within-subjects repeated-measures analysis to address Aim 1 (i.e., the main effect of exercise appointment: cannabis vs non-cannabis). The power analysis indicated that with an alpha of 0.05, $N = 36$ participants would give us a power of 0.80 to detect a small-to-moderate effect size ($f = 0.20$). Based on prior work in the exercise intervention domain [48], and because outcome measures were conducted across several appointments, we estimated an attrition rate of approximately 30%. Thus, we enrolled 52 participants into the study to ensure adequate power at the end of the study. With our final analytic sample of 42 participants, our power to detect these effects is 0.87.

Data collection occurred from August 2021 to November 2022. Participants were compensated up to $100 for their time and effort. The study protocol was approved by the University of Colorado Boulder Institutional Review Board.

2.2 Procedures

This study involved three visits to our exercise facility: (1) a baseline appointment; (2) a cannabis exercise appointment; and (3) a non-cannabis exercise appointment (Fig. 1 of the Electronic Supplementary Material [ESM]). The order in which participants attended each exercise appointment (i.e., cannabis or non-cannabis first) was randomized using a random numbers table generated by a research staff member. Thus, although research staff members who interacted with participants were aware of which exercise appointment it was, they were not aware of what cannabis product participants were assigned to use (i.e., THC or CBD).

Appointments were held at least 48 h apart to allow for proper recovery from exercise. Participants were asked to abstain from cannabis use the day of each appointment (i.e., beginning at 12:00 a.m. the day of their appointment), and were also instructed to refrain from alcohol, caffeine, and tobacco in the 3 h before each appointment. Participants received detailed instructions about how to prepare for their appointment (e.g., recommended fluid intake, caloric intake, and hours of sleep).

2.2.1 Baseline Appointment

Participants who met inclusion criteria were scheduled for a baseline appointment. After providing informed consent, participants completed a baseline questionnaire (see measures below). Additionally, female participants were screened via a urinalysis for pregnancy.

Talk Test

After completing the baseline questionnaire, participants were fitted with a heart rate sensor chest strap (H9; Polar

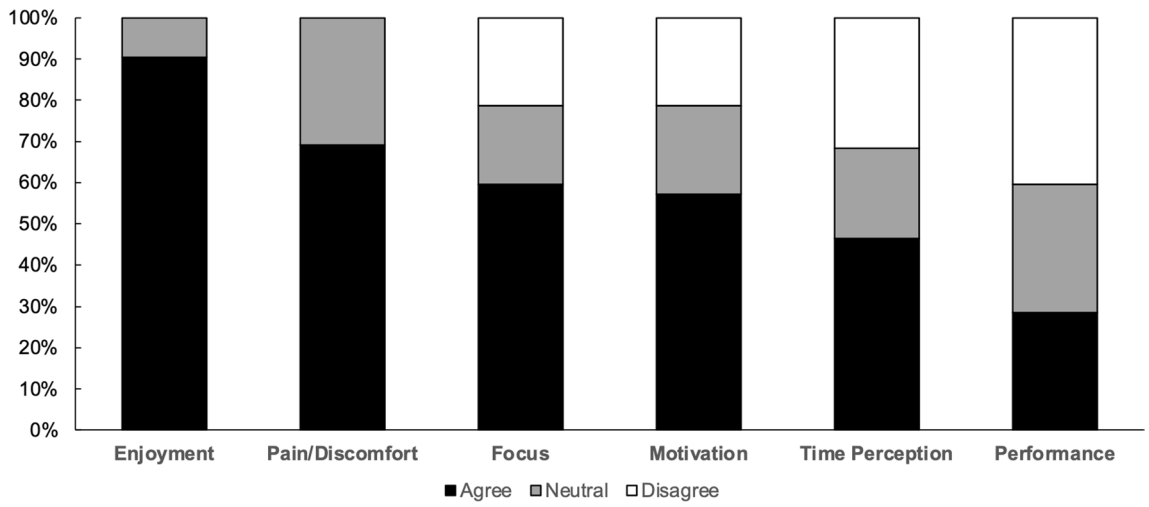

Fig. 1 At baseline, the percentage of participants who agreed, disagreed, or felt neutral toward whether the use of cannabis (1) enhances exercise enjoyment, focus, motivation, and performance, (2) reduces pain/discomfort during exercise, and (3) makes time go by faster during exercise

Electro, Kempele, Finland) prior to undergoing a Talk Test [49] on a treadmill (Trackmaster; Full Vision Inc., Newton, KS, USA) to determine an exercise intensity near their ventilatory threshold (VT). The Talk Test is a widely used method of exercise prescription [50] and has been shown to reliably approximate VT in a variety of populations [49–52]. At the conclusion of the Talk Test, participants practiced responding to the acute exercise measures (see measures below) that had been explained to them in detail prior to the Talk Test. The final speed and incline of the treadmill were recorded, and these parameters were used for the 30-min exercise bout during the cannabis and non-cannabis exercise appointments.

Condition Assignment

After completion of the Talk Test, participants were quasi-randomized to one of two cannabis flower products: a THC-dominant product (24% THC and 1% CBD) or a CBD-dominant product (1% THC and 20% CBD). Notably, as cannabis is classified as a Schedule 1 controlled substance, federal regulations prohibit true random assignment to a cannabis condition in studies of legal-market cannabis use [53]. As such, in line with previous research [54], we employed a quasi-randomization procedure that involved participants rolling a die in which each number on the die represents one of the cannabis conditions, and each cannabis condition has an equal probability of being rolled. As institutional review board regulations preclude us from mandating that participants accept their condition assignment, after being informed of their assigned condition, participants could either accept that assignment or switch to the other condition. Participants were quasi-randomized to a cannabis condition by a member of the research team not otherwise involved in the study.

Participants were instructed to purchase one gram of their assigned product at a local study-partnered dispensary. Consistent with State of Colorado Requirements, the THC and CBD potency of all study products was labeled according to testing in an International Organization of Standards 17025 accredited laboratory. Research team members who ran participant appointments and analyzed participant data were blind to the cannabis condition.

2.2.2 Cannabis Exercise Appointment

A member of the research team traveled to participants' residences in a federally compliant, university-approved mobile pharmacology laboratory [53]. Once the mobile pharmacology laboratory arrived, participants used their assigned cannabis product ad libitum (i.e., according to their typical use patterns) in their residence before returning to the mobile pharmacology laboratory. Specifically, participants were asked to "please use your assigned cannabis product as you normally would during exercise." To estimate how much product participants used during the cannabis exercise appointment, participants weighed their product before and after use on a study-provided scale. Participants also indicated their method of inhalation (e.g., joint, pipe) and the number of puffs/drags taken on a form provided to them.

Participants immediately returned to the mobile pharmacology laboratory after cannabis use, where the

research team member then collected 7 mL of venous blood through venipuncture of a peripheral arm using standard sterile phlebotomy techniques. Quantification of the concentration of cannabinoids (e.g., THC, CBD) was conducted using validated high-performance liquid chromatography-tandem mass-spectroscopy (API5500) in MRM mode [55]. Then, participants were driven in the mobile pharmacology laboratory to the on-campus exercise facility.

Pre-Exercise Procedure

Upon arrival at the exercise facility, participants were fitted with a heart rate sensor chest strap, had their resting heart rate recorded,[1] and completed a brief pre-exercise questionnaire (see measures below). After participants completed the pre-exercise questionnaire, they immediately began the 30-min exercise bout. The number of minutes between acute cannabis use and the start of the 30-min exercise bout was recorded (mean [M] = 32.66 min, standard deviation [SD] = 7.60 min; did not significantly differ by cannabis condition, $p = 0.41$).

30-Minute Exercise Bout

Participants warmed up on the treadmill until they reached the speed and incline determined during their Talk Test. As the Talk Test was individualized for each participant, treadmill speed, incline, and intensity of this exercise bout varied across participants. Once participants reached the pre-determined treadmill speed and incline, the 30-min exercise bout began. Participants' heart rates and responses to the acute exercise measures (see measures below) were recorded four times throughout the exercise bout: at 0, 10, 20, and 30 min. Once the 30-min exercise bout was complete, participants were given the opportunity to cool down.

Post-Exercise Procedure

Participants then completed a brief post-exercise questionnaire (see measures below). At the conclusion of the appointment, a member of the research team ordered a rideshare to return participants to their residence.

2.2.3 Non-Cannabis Exercise Appointment

The procedures for the non-cannabis exercise appointment were the same as the procedures for the cannabis exercise appointment with one exception: participants did *not* use their cannabis product. To maintain uniformity in procedures across the cannabis and non-cannabis exercise appointments, participants were still driven to the exercise facility in the mobile pharmacology laboratory and returned to their residence in a rideshare. Participants also had their blood drawn in the mobile pharmacology laboratory to assess same-day cannabis abstinence. On average, participants completed their cannabis and non-cannabis exercise appointments 9.09 days apart (SD = 5.40), at approximately the same time of day (M = 1.32 h time difference, SD = 2.01).

2.3 Measures

All data were collected using REDCap (https://www.project-redcap.org).

2.3.1 Baseline Questionnaire

Godin Leisure-Time Exercise Questionnaire

The Godin Leisure-Time Exercise Questionnaire [56] was used to assess the frequency and intensity of participants' exercise participation during an average week. Specifically, participants were asked to report the number of minutes per week that they engaged in moderate or vigorous intensity exercise.

Current Cannabis Use

Participants were asked to report the frequency with which they (a) "smoke or vaporize cannabis", (b) "consume cannabis orally or consume edibles", and (c) "use cannabis concentrates (e.g., hash, oil, wax)". For each cannabis form, response options ranged from 1 (I never use [flower/edibles/concentrates]) to 12 (daily).

Cannabis Use Before Exercise

Participants were asked how many times they use cannabis an hour before or during exercise in an average month, as well as the forms of exercise they reported using cannabis in conjunction with most often.

Cannabis-Exercise Attitudes

Attitudes towards using cannabis with exercise were assessed with a six-item measure [30]. Participants were asked what effect they thought acute cannabis use had on exercise-related "enjoyment," "motivation," "ability to focus," "discomfort/pain," and "perception of time". Response options for each item ranged from 1 (makes it much worse) to 5 (makes it much better) (although note that response options for "perception of time" ranged from 1 [makes it much slower] to 5 [makes it much faster]).

[1] We began collecting heart rate data midway through the study. As such, heart rate data are only available for 73.8% of participants (n = 31).

2.3.2 Pre-Exercise Questionnaire

Subjective High

Subjective high was assessed with a single-item slider scale ("I feel high [as in 'drug high'] right now") [57]. Response options ranged from 0 (not at all) to 100 (the strongest feeling possible).

2.3.2.1 Acute Exercise Measures Participants responded to the following measures four times during their cannabis and non-cannabis exercise bouts: at 0 (i.e., once they reached their predetermined treadmill speed/incline and 30-min timing began), 10, 20, and 30 min. At 0 min, participants rated how they felt at that moment. At 10, 20, and 30 min, participants rated how they had felt during the past 10 min. Participants were shown cue cards for each measure and their responses were recorded by a research staff member.

Rate of Perceived Exertion

The Borg Rating of Perceived Exertion Scale [58] was used to assess participants' perceived exertion. Participants were asked to rate their average perceived exertion, i.e., how difficult the exercise feels (at that moment/during the past 10 min) on a 25-point scale that ranged from 6 (no exertion at all) to 20 (maximal exertion).

Pain

The Borg Pain Scale [59] was used to assess participants' pain levels. Participants were asked to indicate how much pain they felt (at that moment/during the past 10 min) on an 11-point scale that ranged from 0 (nothing at all) to 10 (excruciating).

Affective Valence

The Feeling Scale [60] was used to assess participants' affective valence. Participants were asked to indicate how they felt (at that moment/during the past 10 min) on an 11-point scale that ranges from − 5 (very bad) to + 5 (very good).

Affective Arousal

The Felt Arousal Scale [61] was used to assess participants' affective arousal. Participants were asked to indicate how "worked-up" they felt [at that moment/during the past ten minutes] on a 6-point scale that ranged from 1 (low arousal) to 6 (high arousal).

Enjoyment

A single-item measure from Gillman and Bryan [62] was used to assess participants' enjoyment levels. Participants were asked to indicate how enjoyable exercise was [at that moment/during the past ten minutes] on a 5-point scale that ranged from 1 (no enjoyment at all) to 5 (a great deal of enjoyment).

$\%HR_{max}$

To calculate $\%HR_{max}$, participants' heart rate (bpm) at each timepoint was divided by their maximum heart rate, which was determined by subtracting their age from 220.

2.3.3 Post-Exercise Questionnaire

Runner's High Symptoms

A four-item measure from Hicks et al. [36] was used to assess runner's high symptoms. Participants indicated the extent to which they felt four common runner's high symptoms (i.e., "strong," "euphoric," "relaxed," "effortless") during their 30-min exercise bout, on a scale ranging from 1 (not at all) to 5 (extremely). Items were averaged to create a single runner's high symptoms score ($\alpha_{\text{Cannabis}} = 0.65$; $\alpha_{\text{Non-Cannabis}} = 0.79$).

2.4 Planned Analyses

The first aim of the study was to examine the acute effects of using cannabis versus not using cannabis on RPE, pain, affective valence, affective arousal, enjoyment, and runner's high symptoms during the 30-min exercise bout, and the second aim of the study was to examine the acute effects of using a THC- versus CBD-dominant product on the same outcomes. As an exploratory aim, we also examined the acute effects of using cannabis versus not using cannabis on resting heart rate (RHR) before the 30-min exercise bout and percentage of maximal heart rate ($\%HR_{max}$) during the 30-min exercise bout.

These analyses involved two *within-subjects* factors[2] (exercise appointment: cannabis vs non-cannabis, and time: four repeated measures of RPE, pain, affective valence, affective arousal, enjoyment, $\%HR_{max}$) and one *between-subjects* factor (cannabis condition: THC vs CBD). To conduct these analyses, we ran separate mixed-effects models (one for each dependent variable: RPE, pain, affective valence, affective arousal, enjoyment, runner's high symptoms, RHR, and $\%HR_{max}$) estimating random intercepts

[2] Runner's high symptoms and RHR were only assessed at one timepoint (runner's high symptoms: post-exercise; RHR: pre-exercise). As such, these models only included one within-subjects factor (i.e., exercise appointment [cannabis vs. non-cannabis] but not time) and estimated random intercepts (but not slopes) for participant. These models therefore examined (1) average runner's high symptoms/RHR, (2) the effect of exercise appointment on runner's high symptoms/RHR, (3) the effect of cannabis condition on runner's high symptoms/RHR, and (4) the exercise appointment by cannabis condition interaction.

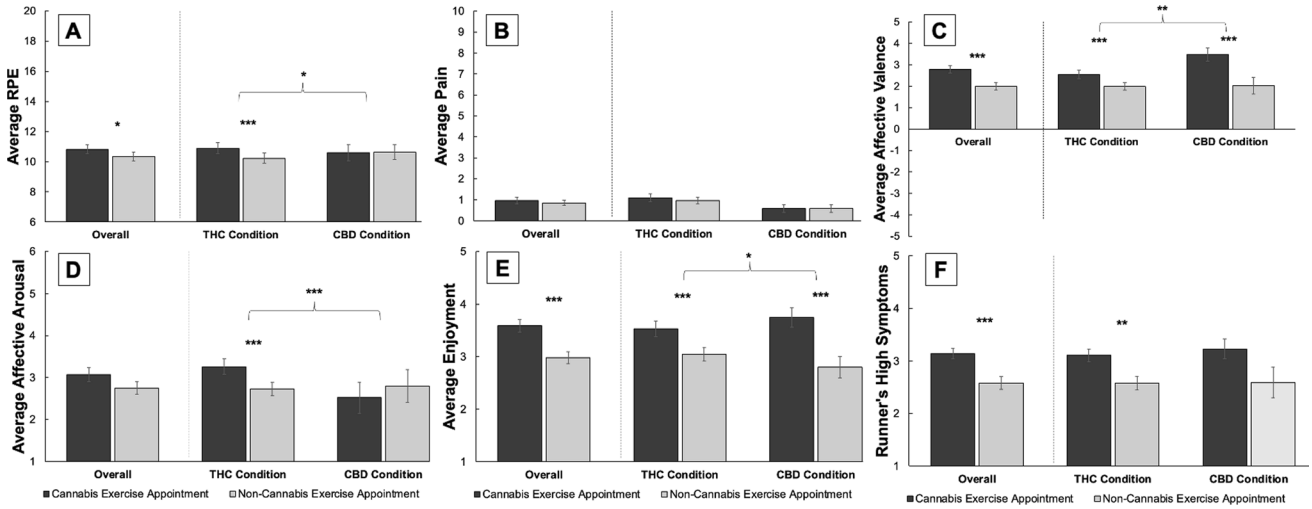

Fig. 2 Average (**A**) ratings of perceived exertion (RPE), (**B**) pain, (**C**) affective valence, (**D**) affective arousal, (**E**) enjoyment, and (**F**) runner's high symptoms by exercise appointment and cannabis condition. Note: response options for each measure ranged from: 6–20 (RPE), 0–10 (pain), − 5 to + 5 (affective valence), 1–6 (affective arousal), 1–5 (enjoyment), and 1–5 (runner's high symptoms). *CBD* cannabidiol, *THC* Δ9-tetrahydrocannabinol

and slopes[2] for participant. Models examined (1) average levels of the dependent variable (e.g., what is the average level of RPE collapsing across time, exercise appointment, and cannabis condition?), (2) linear change of the dependent variable over time (e.g., across exercise appointment and cannabis condition, is there a linear increase in RPE?), (3) quadratic change of the dependent variable over time (e.g., across exercise appointment and cannabis condition, is there a quadratic increase in RPE?), (4) the interactions of these factors with exercise appointment, which was contrast coded to compare the effects of using versus not using cannabis prior to the exercise bout (e.g., across the THC and CBD conditions, does RPE differ between the cannabis and non-cannabis exercise appointment?), (5) the interactions of these factors with cannabis condition, which was contrast coded to compare the effects of using a CBD-versus THC-dominant product (e.g., across the cannabis and non-cannabis exercise appointment, does RPE differ between participants assigned to use a THC-dominant product and participants assigned to use a CBD-dominant product?), and (6) the interactions of these factors with exercise appointment and cannabis condition (e.g., does RPE vary as a function of both cannabis condition and exercise appointment?). Simple effects tests were conducted to explore interactions (*Note*. For simple effects tests, a Bonferroni-corrected significance level of $\alpha = 0.05/16 = 0.0031$ was used to adjust for multiple comparisons).

Analyses were conducted in R version 4.1.2 (www.rstudio.com). Mixed-effects models were conducted using the lme4 package [63], which implements maximum likelihood estimation. All beta coefficients (*B*) reported in the results below are unstandardized.

3 Results

Fifty-two participants were recruited for the study, of whom 44 completed all study components. Two further participants were excluded from analyses resulting in a final sample of 42 participants (THC condition = 31, CBD condition = 11; Fig. 2 of the ESM). Although those who attritted were marginally more likely to have been assigned to the CBD condition ($p = 0.06$), no other significant differences in demographics or baseline characteristics emerged between participants who did versus did not complete the study. See Table 1 for demographics and baseline characteristics overall and by condition.

At baseline, most participants reported that cannabis increased their enjoyment of exercise (90.5%), reduced their pain/discomfort levels during exercise (69.0%), improved their ability to focus during exercise (59.5%), and increased their motivation to exercise (57.1%). Only 45.2% reported that cannabis made time go by faster during exercise, and only 28.6% reported that cannabis improved their exercise performance (Fig. 1). The forms of exercise that participants reported using cannabis in combination with most often were running/jogging ($n = 32$), hiking ($n = 24$), and biking ($n = 17$).

3.1 Manipulation Check

Participants in the THC condition displayed significantly higher plasma THC concentrations (B = 137.73, standard error [SE] = 53.83, $p = 0.01$) and significantly lower plasma CBD concentrations (B = − 32.79, SE = 4.72, $p < 0.001$)

Table 1 Participant demographics and baseline characteristics

	Overall $N=42$	THC $n=31$	CBD $n=11$	*p* value
Demographics				
Age	30.81 (4.72)	31.16 (4.77)	29.82 (4.64)	0.42
Gender				0.67
Male	28 (66.7%)	20 (64.5%)	8 (72.7%)	
Female	12 (28.6%)	9 (29.0%)	3 (27.3%)	
Non-binary	2 (4.6%)	2 (6.5%)	0 (0.0%)	
Sexual orientation				0.77
Straight/heterosexual	23 (79.3%)	17 (77.3%)	6 (85.7%)	
Lesbian or gay	1 (3.4%)	1 (4.5%)	0 (0.0%)	
Bisexual	3 (10.3%)	2 (9.1%)	1 (14.3%)	
Not sure	2 (6.9%)	2 (9.1%)	0 (0.0%)	
Ethnicity				0.70
Hispanic	3 (7.1%)	3 (9.7%)	0 (0.0%)	
Non-Hispanic	39 (92.9%)	28 (90.3%)	11 (100.0%)	
Race				
White	37 (88.1%)	27 (87.0%)	10 (90.9%)	0.67
Asian	3 (7.1%)	2 (6.5%)	1 (9.1%)	
Two or more races	2 (4.8%)	2 (6.5%)	0 (0.0%)	
Education				0.29
High school/GED	4 (9.5%)	3 (9.7%)	1 (9.1%)	
Some college	2 (4.8%)	2 (6.5%)	0 (0.0%)	
Bachelor's degree	23 (54.8%)	18 (58.1%)	5 (4.5%)	
Master's degree	10 (23.8%)	5 (1.6%)	5 (4.5%)	
Doctoral degree	3 (7.1%)	3 (9.7%)	0 (0.0%)	
Employment status				0.69
Full-time	29 (69.0%)	21 (67.7%)	8 (72.7%)	
Part-time	8 (19.0%)	7 (22.6%)	1 (9.1%)	
Unemployed	3 (7.1%)	2 (6.5%)	1 (9.1%)	
Full-time student	2 (4.8%)	1 (3.2%)	1 (9.1%)	
Baseline characteristics				
Cannabis frequency				
Flower	8.41 (3.90)	8.60 (4.00)	7.91 (3.78)	0.62
Edibles	4.50 (3.01)	4.55 (3.13)	4.36 (2.77)	0.86
Concentrates	3.52 (4.03)	3.32 (3.66)	4.09 (5.09)	0.59
Cannabis use disorder score	2.07 (2.22)	2.42 (2.35)	1.09 (1.51)	0.09
Exercise minutes/week	382.95 (429.12)	408.68 (479.28)	310.45 (240.90)	0.52
L-Cat	4.88 (0.97)	4.97 (0.91)	4.64 (1.12)	0.34
Cannabis before exercise/months	9.10 (9.80)	9.32 (10.27)	8.45 (8.76)	0.80

GED General Equivalency Diploma, *L-Cat* Stanford Leisure-Time Activity Categorical Item

Note: mean (standard deviation) or *n* (%). We began collecting data on sexual orientation midway through the study. As such, sexual orientation data are only available for 69.0% of participants ($n=29$). Responses to cannabis frequency items range from 1 (I never use [flower/edibles/concentrates]) to 12 (daily). Average cannabis flower frequency scores corresponded to using approximately 3–4 days per week, average cannabis edible frequency scores corresponded to using approximately 2–3 days per month, and average cannabis concentrates scores corresponded to using approximately 1–2 days per month. Cannabis use disorder scores [78] range from 0 to 10, with higher scores indicating more cannabis dependence. Scores on the Alcohol Use Disorder Test (AUDIT-C [79]) range from 0 to 12, with higher scores indicating more alcohol dependence. Responses to the L-Cat [80] range from 1 (very inactive) to 6 (very active). Cannabis before exercise/months refers to the number of times per month participants report using cannabis an hour before and/or during exercise

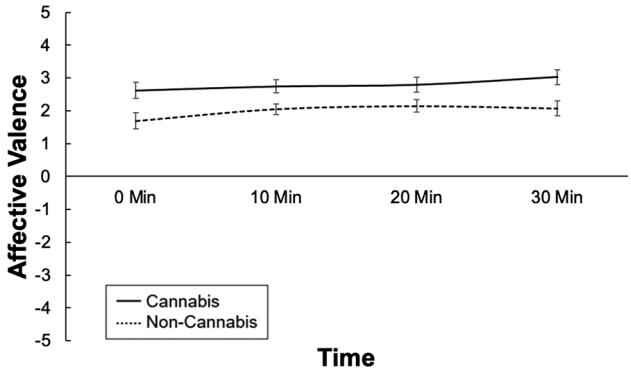

Fig. 3 Affective valence over time by exercise appointment. Note: affective valence response options ranged from 1 to 6. See Fig. 4 of the ESM for ratings of perceived exertion, pain, affective arousal, and enjoyment over time by exercise appointment. *Min* minutes

during their cannabis exercise appointment than participants in the CBD condition, suggesting that participants complied with their assigned condition. Additionally, participants displayed significantly higher plasma THC (B = 103.74, SE = 31.73, $p = 0.002$) and CBD (B = 7.21, SE = 3.67, $p = 0.05$) concentrations during their cannabis versus non-cannabis exercise appointment (Fig. 3 of the ESM). See Table 2 for information regarding ad libitum cannabis use during the cannabis exercise appointment.

3.2 Subjective Exercise Response

Average RPE, pain, affective valence, affective arousal, enjoyment, and runner's high symptoms during the 30-min exercise bout by exercise appointment and condition (THC vs CBD) are presented in Fig. 2. Although RPE, pain, affective valence, affective arousal, and enjoyment were recorded four times throughout the exercise bout (Fig. 3; Fig. 4 of the ESM), there were no significant appointment by time ($ps > 0.10$) or condition by time ($ps > 0.20$) interactions. As such, we focus on the main effect of exercise appointment and cannabis condition collapsing across timepoint below. On average, participants ran at a pace of 8.79 km per hour (SD = 1.83) with an incline of 1.81% (SD = 2.92) during the two 30-min exercise bouts. Speed and incline did not differ significantly between those in the THC and those in the CBD condition ($ps = 0.66$).

RPE

On average, participants reported significantly more exertion during their cannabis (vs non-cannabis) exercise appointment (B = 0.32, SE = 0.15, $p = 0.04$). This effect was moderated by cannabis condition (i.e., significant exercise appointment X cannabis condition interaction; B = 0.68, SE = 0.30, $p = 0.02$), such that there was a larger difference in exertion between the cannabis and non-cannabis exercise appointment among participants in the THC (vs CBD) condition. Simple effects tests indicated that participants in the THC ($p < 0.001$) condition but not the CBD ($p = 0.93$) condition reported more exertion during their cannabis (vs non-cannabis) exercise appointment.

Pain

On average, participants' pain levels did not significantly differ between the cannabis and non-cannabis exercise appointment (B = 0.07, SE = 0.09, $p = 0.45$). This effect was not moderated by cannabis condition (B = 0.14, SE = 0.18, $p = 0.45$), and simple effect tests indicated that participants in both the THC ($p = 0.14$) and CBD ($p = 0.99$) conditions reported similar (low) levels of pain during both exercise appointments.

Table 2 Cannabis use during the cannabis exercise appointment

	Overall $N = 42$	THC $n = 31$	CBD $n = 11$	*p* value
Grams used	0.37 (0.32)	0.36 (0.34)	0.39 (0.29)	0.82
Method of inhalation				0.05
Joint	19 (45.2%)	15 (48.4%)	4 (36.4%)	
Pipe	12 (28.6%)	9 (29.0%)	3 (27.3%)	
Bong	8 (19.0%)	7 (22.6%)	1 (9.1%)	
Vaporizer	2 (4.8%)	0 (0.0%)	2 (18.2%)	
Bubbler	1 (2.4%)	0 (0.0%)	1 (9.1%)	
Number of drags/puffs	10.90 (10.49)	10.00 (8.89)	13.70 (14.63)	0.34
Plasma THC (ng/mL)	87.6 (121)	114 (131)	12.7 (11.6)	0.04
Plasma CBD (ng/mL)	10.7 (24.4)	0.5 (0.5)	40.0 (34.9)	< 0.001
Subjective high	66.81 (21.06)	73.71 (13.76)	47.36 (26.22)	< 0.001

CBD cannabidiol, *THC* Δ9-tetrahydrocannabinol

Note: subjective high response options ranged from 0 (not at all) to 100 (the strongest feeling possible)

Affective Valence

On average, participants reported higher affective valence (i.e., more positive affect) during their cannabis (vs non-cannabis) exercise appointment (B = 1.01, SE = 0.15, $p < 0.001$). This effect was moderated by cannabis condition (B = −0.88, SE = 0.29, $p = 0.003$), such that there was a larger difference in affective valence between the cannabis and non-cannabis exercise appointment among participants in the CBD (vs THC) condition. Simple effect tests indicated that participants in both the THC ($p < 0.001$) and CBD ($p < 0.001$) conditions reported higher affective valence during their cannabis (vs non-cannabis) appointment.

Affective Arousal

On average, participants' affective arousal did not significantly differ between their cannabis and non-cannabis exercise appointment (B = 0.13, SE = 0.11, $p = 0.23$). This effect was moderated by cannabis condition (B = 0.80, SE = 0.21, $p < 0.001$), such that there was a larger difference in affective arousal between the cannabis and non-cannabis exercise appointment among participants in the THC (vs CBD) condition. Simple effect tests indicated that participants in the THC ($p < 0.001$) but not the CBD ($p = 0.13$) condition reported more affective arousal during their cannabis (vs non-cannabis) exercise appointment.

Enjoyment

On average, participants reported significantly more enjoyment during their cannabis (vs non-cannabis) exercise appointment (B = 0.72, SE = 0.10, $p < 0.001$). This effect was moderated by cannabis condition (B = −0.47, SE = 0.20, $p = 0.02$), such that there was a larger difference in enjoyment between the cannabis and non-cannabis exercise appointment among participants in the CBD (vs THC) condition. Simple effect tests indicated that participants in both the THC ($p < 0.001$) and CBD ($p < 0.001$) conditions reported more enjoyment during their cannabis (vs non-cannabis) exercise appointment.

Runner's High Symptoms

On average, participants reported experiencing more runner's high symptoms during their cannabis (vs non-cannabis) exercise appointment (B = 0.58, SE = 0.14, $p < 0.001$). This effect was not moderated by cannabis condition (B = −0.10, SE = 0.29, $p = 0.72$), although simple effects tests indicated that participants in the THC ($p = 0.001$) condition but not the CBD ($p = 0.01$) condition reported experiencing more runner's high symptoms during their cannabis (vs non-cannabis) exercise appointment.

3.3 Heart Rate

See Fig. 5 of the ESM for participants' absolute heart rates before and during exercise.

RHR

Participants' RHR was recorded upon arrival at the exercise facility, and as such, during the cannabis exercise appointment, was recorded after participants had used their cannabis product. On average, participants' RHR was higher during their cannabis (vs non-cannabis) exercise appointment (B = 5.76, SE = 2.58, $p = 0.03$). This effect was marginally moderated by cannabis condition (B = 9.52, SE = 5.15, $p = 0.07$), and simple effect tests indicated that participants in the THC ($p < 0.001$) condition but not the CBD ($p = 0.82$) condition had a higher RHR prior to exercising during their cannabis (vs non-cannabis) exercise appointment.

$\%HR_{max}$

On average, participants' $\%HR_{max}$ did not significantly differ between their cannabis and non-cannabis exercise appointment (B = 1.35, SE = 0.92, $p = 0.15$). This effect was not moderated by cannabis condition (B = 1.22, $SE = 1.85$, $p = 0.51$), and simple effect tests indicated that $\%HR_{max}$ did not significantly differ between the cannabis and non-cannabis exercise appointment among participants in the THC ($p = 0.04$) condition or CBD ($p = 0.64$) condition.

4 Discussion

Despite increasing public interest in the use of cannabis with exercise [12–19], there is limited empirical evidence regarding the potential harms and benefits of this behavior. Using a federally compatible at-home administration procedure in concert with a mobile pharmacology laboratory, this is the first study to investigate the acute effects of commercially available cannabis on subjective responses to exercise in a controlled laboratory environment. Moreover, this is the first investigation to assign participants to use either a THC-dominant or CBD-dominant cannabis product in the context of exercise, thus providing novel insights into the effect of cannabinoid content on responses to exercise. Although acute cannabis use led to a more positive subjective exercise experience in terms of affective valence, enjoyment, and runner's high symptoms, it also led to greater exertion in our sample of regular cannabis users. Interestingly, these effects appeared to depend, in part, on cannabinoid content, with participants in the THC condition reporting more exertion and affective arousal, and participants in the CBD condition reporting more enjoyment and positive affect.

Across both the THC and CBD conditions, participants reported more positive affect and enjoyment while running under the influence of cannabis, which is in line with recent reports of cannabis being used to facilitate exercise enjoyment [10, 29]. Of note, cannabis use appeared to have a stronger impact on positive affect and enjoyment among participants in the CBD (vs THC) condition. This finding was surprising, given that CBD is non-psychoactive [42] and THC intoxication has been associated with increased positive affect among regular cannabis users [39]. However, there is some indication that CBD, without the presence of THC, may improve affective responses to exercise, as demonstrated in a recent study that administered 300 mg of isolated CBD during a 1-h treadmill session [64]. Moreover, as our ad libitum administration procedure allowed participants to self-titrate to their desired high, it is important to acknowledge that participants in the CBD condition still reported feeling high and had detectable concentrations of plasma THC during their cannabis exercise appointment, albeit to a lesser degree than participants in the THC condition. As the effects of THC on psychotomimetic effects (e.g., anxiety, paranoia) are dose dependent [45], it is therefore possible that participants in the CBD condition experienced fewer unmeasured negative effects than participants in the THC condition, resulting in a more positive subjective exercise experience for participants using a CBD-dominant product. Future studies should assess individuals' perceptions of anxiety, paranoia, and other psychotomimetic effects while exercising after cannabis use.

Although acute cannabis use was associated with a more positive exercise experience in terms of affective valence, runner's high symptoms, and enjoyment, it was also associated with increased perceptions of exertion among participants in the THC condition. This finding was surprising, given that an exercise-induced positive affect has been associated with reductions in exertion during exercise [41]. However, it is in line with previous research documenting an ergolytic effect of cannabis use on exercise performance [24, 25], as well as participants' perceptions at baseline that cannabis would either hinder or have no impact on their exercise performance. Further, it is in line with research suggesting that THC intoxication may lead to feelings of tiredness [43].

Participants reported similar levels of pain during both their cannabis and non-cannabis exercise appointments, which raises the possibility that the analgesic effects of cannabis may not generalize to the context of exercise. This finding differs from previous research suggesting that pain management is a common reason for using cannabis before exercise [65], and contradicts a previous study that found individuals reported lower levels of pain after running with, versus without, cannabis [30]. Several methodological differences may explain this discrepancy. First, as Gibson and Bryan's [30] study involved free-living exercise, participants reported running an average of 19 s per kilometer slower after acute cannabis use, which could have led them to experience less pain during their cannabis run. Second, participants in Gibson and Bryan's [30] study reported their pain levels *after* the conclusion of their run, rather than during, which was the case in the present study. As such, it is possible that cannabis use may reduce pain experienced during exercise recovery, rather than pain experienced during acute exercise. This could explain why many individuals report using cannabis to assist with recovery from exercise [10]. Regardless, it is important to note that average pain levels were very low during both the cannabis (0.96 out of ten) and non-cannabis (0.86 out of ten) exercise appointments, and as a result, the influence of cannabis on exercise-induced pain in this study could have been masked by a floor effect. Future studies are needed to examine whether findings generalize to populations who experience more exercise-induced pain, such as individuals with arthritis or older adults.

Heart rate was elevated after ad libitum cannabis use among participants in the THC condition but not the CBD condition, in line with previous research documenting an increase in heart rate following inhalation of THC [66] but not CBD [67]. Importantly, the magnitude of this increase did not appear to differ between the pre-exercise and acute-exercise timepoints, suggesting that THC and acute exercise do not have a *compounding* effect on the heart rate. As physiological responses to exercise (e.g., heart rate) are strongly tied to subjective responses to exercise [68], this may be why participants in the THC condition, but not the CBD, condition reported more perceived exertion and affective arousal while running after ad libitum cannabis use.

Of note, to maximize external validity, the cannabis products employed in this study were commercially available and widely used. Participants purchased their products from a study-partnered dispensary, therefore enhancing the study's ecological validity. This addresses a major limitation of previous work in the cannabis and exercise domain, which has mostly examined the acute effects of cannabis products that bear little resemblance in THC potency or quality to cannabis used in state-regulated markets today [23–27]. Further, as participants were instructed to use their products according to their typical use patterns, we received important insights into the route of consumption (e.g., joint, pipe) and amount of cannabis that individuals may gravitate toward in the context of exercise.

The inclusion of a Talk Test at baseline allowed us to personalize participants' exercise bouts according to their individual fitness levels. As such, although participants were all exercising at a different pace, their $\%HR_{max}$ indicated that they were all exercising near their VT [69, 70]. This approach allowed us to hold the intensity of exercise (i.e.,

speed, incline) constant between the cannabis and non-cannabis exercise appointments, overcoming a key limitation of Gibson and Bryan [30], which saw participants running an average of 19 s/km slower during their cannabis (vs non-cannabis) run. This suggests that the higher levels of both positive (e.g., affect, enjoyment) and negative (e.g., exertion) subjective effects observed during the cannabis exercise appointment were driven by acute cannabis use rather than a difference in running pace.

4.1 Study Limitations

Findings should be interpreted in light of several limitations. Notably, we were unable to include a placebo-control condition because of the lack of such products in today's legal market as well as state regulations requiring that THC and CBD content be labeled on all legal market cannabis products. Federal cannabis regulations also prohibited us from employing a standardized administration procedure, which further limited our ability to pinpoint causal effects. Additionally, as federal regulations prohibit the consumption of state-regulated, commercially available cannabis products in a laboratory setting [53], participants had to administer their assigned cannabis product ad libitum in their own residence before being driven to the exercise facility in the mobile pharmacology laboratory, leading to average delay of 32 min between cannabis consumption and the cannabis exercise bout.

Although the decision to quasi-randomize participants to use either a THC-dominant or CBD-dominant product allowed us to explore whether any of the acute effects of cannabis on exercise varied as a function of cannabinoid content (i.e., THC:CBD) these findings should be interpreted with caution given the small number of participants in each condition. Moreover, as institutional review board regulations prohibited us from mandating that participants accept their condition assignment, at baseline, several participants assigned to the CBD condition elected to switch to the THC condition. This quasi-randomization, in tandem with a slightly higher attrition rate in the CBD condition, led to an unbalanced sample and suggests that participants may prefer to use THC products in the context of exercise, in line with previous research [30]. Future studies may wish to employ a crossover design in which all participants exercise without cannabis, with THC, and with CBD to increase power and rule out the possibility of between-subject confounding variables.

Although the research team was blind to the cannabis condition, participants were not; state legislation requires that THC and CBD content be labeled on all commercially available cannabis products, and as such, participants were aware of the cannabinoid content of their assigned product, which may have increased expectancy effects, the likelihood of which was perhaps exacerbated by the fact that outcome measures were subjective in nature. Indeed, participants reported favorable attitudes toward cannabis use with exercise at baseline (e.g., 90.5% reported that cannabis increased their enjoyment of exercise). However, as participants all had previous experience using cannabis in the context of exercise, it is unclear whether these findings reflect expectancy effects or previous experience.

As with all laboratory studies, our study had trade-offs in terms of ecological validity; participants were running in a laboratory, on a treadmill, at a prescribed pace and incline, in front of research assistants. As such, it is likely that the context of exercise in this study did not reflect participants' typical exercise context when using cannabis in the real world (e.g., outdoors [29, 30]). Additionally, average $\%HR_{max}$ over the course of the cannabis (75.03% HR_{max}) and non-cannabis (73.38% HR_{max}) exercise bouts suggested that most participants were engaging in moderate-to-vigorous intensity exercise [71]. As the subjective experience of exercise has been shown to vary across exercise intensities [72], future studies should examine whether exercise intensity moderates the effects of cannabis on subjective responses to exercise. Relatedly, although the Talk Test allowed us to estimate an exercise intensity near participants' VT, future studies should consider implementing a VO_2 max test for a more precise prescription of exercise intensity (i.e., as a percentage of maximal oxygen uptake [73]). Of note, although this study involved running, participants reported using cannabis with a variety of exercise modalities at baseline and previous research suggests that individuals use cannabis in the context of both aerobic and anaerobic exercise [29]. Thus, more research into the potential negative and positive consequences of cannabis use in the context of different exercise modalities and intensities is needed before any recommendations can be made.

Importantly, this was a *highly* active sample of regular cannabis users. Given safety considerations and institutional review board guidelines prohibiting the enrollment of cannabis-naïve participants in observational studies of exercise during the use of legal-market cannabis, participants needed to report previous experience using cannabis in combination with running or jogging. This raises the possibility of selection effects (i.e., individuals with more positive experiences of cannabis use with exercise may have been more likely to enroll in our study). Relatedly, as this study involved prolonged exercise near VT, participants needed to be physically capable of safely engaging in moderate-to-vigorous intensity exercise for 30 or more minutes. Participants were therefore all healthy habitual exercisers, reporting an average of 383 min of moderate-to-vigorous intensity exercise per week, which is well above minimum guidelines [74].

Additionally, plasma THC concentrations were relatively high during the cannabis exercise appointment for participants in the THC condition (M = 148 ng/mL), which may reflect some degree of tolerance to cannabis impairment in our sample of regular cannabis users. As a result, our study may not be representative of the experiences of all individuals who use cannabis before exercise, and we caution the generalization of our findings to non-users, infrequent cannabis users, cannabis users who have never exercised under the influence of cannabis, or less active populations.

Additionally, although participants' blood was drawn in the mobile pharmacology laboratory immediately after cannabis use, given the long terminal half-life of cannabis [75], it is possible that participants' plasma cannabinoid concentrations may have been influenced by the use of cannabis outside of study times. To assess this possibility, future studies should consider employing a detailed timeline follow-back of cannabis use in the days preceding study appointments (e.g., [76]), as well as additional blood draws both before and after acute cannabis use. Future studies should also consider assessing plasma concentrations of minor cannabinoids (e.g., cannabigerol [CBG], cannabinol [CBN]), given recent evidence in support of the cannabis "entourage effect", which suggests that the interaction of various compounds within the cannabis plant (i.e., THC, CBD, minor cannabinoids, and terpenes) may produce a more significant or unique effect compared with compounds such as THC or CBD alone [77]. Therefore, it is possible that findings from our study may not generalize to studies using isolated forms of THC and CBD, as our study involved ad libitum administration of legal-market cannabis flower, which has been associated with measurable increases in plasma concentrations of minor cannabinoids [38].

Although participants reported their cannabis and alcohol use patterns at baseline, we do not have additional information regarding their use of tobacco, prescription medications, or other substances that may have influenced their responses to cannabis or exercise. Future studies in this domain should consider including a detailed substance use battery at baseline, as well as administering a urine toxicology screen to assess abstinence from illicit substances. Further, a considerable portion of the sample consisted of non-Hispanic White men between the ages of 21 and 40 years. This is in line with previous survey research that has found that individuals who use cannabis with exercise tend to be younger, White, and male [10, 65]. Additionally, the limited age range of our sample was likely due, in part, to the fact that we were unable to enroll men above the age of 40 years and women above the age of 50 years into the study (because of safety concerns related to cardiovascular health). Indeed, we note that a considerable number of individuals above the maximum age requirement inquired about participation but did not meet inclusion criteria based on their age. Future studies should make active efforts to recruit more diverse samples in terms of race/ethnicity, age, gender, and geographic location.

5 Conclusions

The present study addresses calls for more research on how acute cannabis use influences exercise [22, 28, 29] and is the first study, to our knowledge, to examine the acute effects of commercially available cannabis products on subjective responses to exercise in a laboratory environment. This marks an important first step in a nascent field that has, to date, consisted primarily of cross-sectional surveys and studies examining the acute effects of cannabis on exercise and sports *performance*. Our findings suggest that, among individuals who have experience using cannabis with exercise, smoking or vaporizing cannabis flower prior to exercise may lead to increases in both positive (e.g., affect, enjoyment) and negative (e.g., exertion) aspects of the exercise experience. As recreational exercisers tend to use cannabis for non-performance-enhancing reasons (e.g., enjoyment, pain management) [10, 29, 65] and generally agree that cannabis does not improve their exercise performance [10], continued research into the impact of cannabis on the subjective experience of exercise — both positive and negative — is critical.

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Declarations

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Conflicts of Interest/Competing Interests Laurel P. Gibson, Gregory R. Giordano, L. Cinnamon Bidwell, Kent E. Hutchison, and Angela D. Bryan have no conflicts of interest that are directly relevant to the content of this article.

Ethics Approval Approval was obtained from the Ethics Committee of the University of Colorado Boulder (Protocol #19-0596; approval date: 01/22/20). The procedures used in this study adhere to the tenets of the Declaration of Helsinki.

Consent to Participate Informed consent was obtained from all individual participants included in the study.

Consent for Publication Not applicable.

Availability of Data and Material The datasets analyzed during the current study are available from the corresponding author on reasonable request.

Code Availability The R code is available by contacting the corresponding author.

Authors' Contributions LPG, LCB, KEH, and ADB designed the research; LPG and GRG performed the research; LPG analyzed the data; LPG wrote the paper; GRG, LCB, KEH, and ADB provided feedback on the paper. All authors read and approved the final version.

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